Raunaq Singh

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Education

Georgia Institute of Technology August 2026 – May 2029
B.S. in Industrial Engineering, AI/OR for Decisions; Minor in Computing & Theory *Atlanta, GA*

University of Illinois Urbana-Champaign August 2025 – May 2026
Grainger College of Engineering (transferred) *Urbana, IL*

Coursework: Linear Algebra, Discrete Mathematics, Intro to Computing (Python), Data Science Discovery, Data Science Exploration, Calculus I–II

Experience

Software Engineering Intern June 2026 – September 2026
Miraphoto *New York City, NY*

- Shipped a live Organization Profile and Event RSVP portal for a photo-booking app by rebuilding its full UI from a single design screenshot, with role-based access for 3 user types and validated phone, email, and URL fields
- Connected the portal to the live Node/MongoDB API on production and staging by fixing field mismatches found in backend code, and wrote the backend validation schema for organization accounts
- Kept the feature deployable across 3 hosting setups by migrating Next.js to Astro with GitHub Actions, then documented a handoff plan so deployment no longer depends on an intern's personal account

Projects

PokéLab | Live App | Code November 2026

- Built and deployed a public web app that groups all 1,025 Pokémon by stat profile and builds balanced 6-member teams, using a cached, validated data pipeline over PokéAPI and Smogon
- Tested whether Pokémon fall into stable stat groups by bootstrap-testing K-Means and GMM, and found they do not (best ARI 0.59 vs. a 0.60 stability bar), avoiding a misleading result
- Beat greedy picks and 10,000 random teams (only 4% met all rules) by building a team optimizer that models each pick as an integer program over type coverage and balance

Mean-Reversion Pairs Trading | Code September 2026 – October 2026

- Built a pairs trading backtester to test whether price spreads between related stocks revert, screening 8 candidate pairs with Engle-Granger cointegration tests and Benjamini-Hochberg correction
- Removed look-ahead bias and 4 other backtest flaws by adding cointegration tests and a train/test split, so each simulated trade uses only data available at that point
- Found 1 significant pair (KO-PEP, adj. $p = 0.036$) but no reliable profit after trading costs and walk-forward tests, showing the edge does not hold up in realistic conditions

Leadership

Co-Founder June 2022 – August 2024
Community Swim *Hillsborough, NJ*

- Taught swimming and water safety to 15 children ages 4–12 by co-founding a summer swim program
- Prepared 10 of 15 students to join a swim team by designing lessons tailored to each child's skill level

Varsity Swimmer; Team Captain (2024–2025) November 2021 – February 2025
Hillsborough High School Swim Team *Hillsborough, NJ*

- Selected team captain for the 2024–2025 season after competing on varsity for four years
- Led and mentored teammates at practices and meets, keeping a supportive team environment

Technical Skills

Languages: Python, TypeScript, JavaScript
Data & ML: pandas, NumPy, scikit-learn, SciPy, statsmodels, Matplotlib, seaborn, Plotly
Optimization & Apps: PuLP (integer programming), Streamlit
Web Development: MongoDB, Node.js, Next.js, React, Astro, Tailwind CSS, Zod
Tools: Git, GitHub Actions, pytest, Vercel, Jupyter
Interests: Swimming, Cycling, Fishing, Grilling